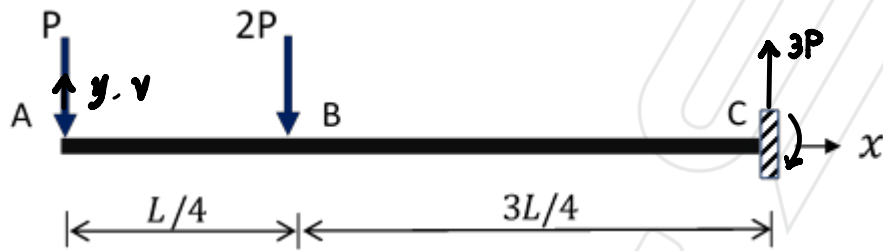


Structures I

Tutorial Sheet 5B: Elastic Curves

1. Determine the elastic curve for the beam shown below.



$$\therefore \frac{d^2v}{dx^2} = -\frac{1}{EI} [-Px - 2P\langle x - \frac{L}{4} \rangle] = -\frac{M}{EI} = -\frac{(-Px - 2P\langle x - \frac{L}{4} \rangle)}{EI}$$

$$\therefore \frac{dv}{dx} = \int \frac{(-Px - 2P\langle x - \frac{L}{4} \rangle)}{EI} dx = \frac{P}{EI} (-\frac{1}{2}x^2 - \langle x - \frac{L}{4} \rangle^2 + C_1)$$

$$\therefore v = \int \frac{P}{EI} (-\frac{1}{2}x^2 - \langle x - \frac{L}{4} \rangle^2 + C_1) dx$$

$$= \frac{P}{EI} (-\frac{1}{6}x^3 - \frac{1}{3}\langle x - \frac{L}{4} \rangle^3 + C_1x + C_2)$$

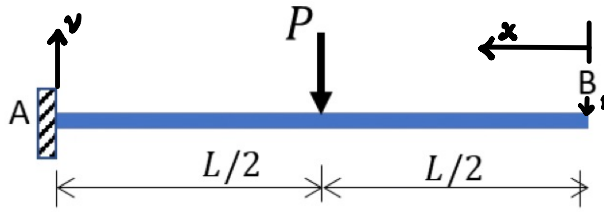
When $x = L$, $\frac{dv}{dx} = 0$ and $v = 0$

$$\therefore C_1 = -\frac{17}{16}L^2$$

$$\therefore C_2 = -\frac{145}{192}L^3$$

$$\therefore v(x) = \frac{P}{EI} (-\frac{1}{6}x^3 - \frac{1}{3}\langle x - \frac{L}{4} \rangle^3 - \frac{17}{16}L^2x - \frac{145}{192}L^3)$$

2. Calculate the maximum deflection using both double integration and unit load approaches. Sketch the deflected shape.



a) $M = -P \langle x - \frac{L}{2} \rangle$

$$\textcircled{1} \frac{dv^2}{dx^2} = -\frac{1}{R} = -\frac{M}{EI} = \frac{P}{EI} \langle x - \frac{L}{2} \rangle$$

$$\therefore \frac{dv}{dx} = \int \frac{P}{EI} \langle x - \frac{L}{2} \rangle dx = \frac{P}{EI} (\frac{1}{2} \langle x - \frac{L}{2} \rangle^2 + C_1)$$

$$\therefore v = \frac{P}{EI} \int (\frac{1}{2} \langle x - \frac{L}{2} \rangle^2 + C_1) dx = \frac{P}{EI} (\frac{1}{6} \langle x - \frac{L}{2} \rangle^3 + C_1 x + C_2)$$

When $x = 0$, $v = 0$ and $\frac{dv}{dx} = 0$

$$\therefore C_1 = -\frac{1}{8}L^2$$

$$\therefore C_2 = \frac{5}{48}L^3$$

$$\therefore v = \frac{P}{EI} (\frac{1}{6} \langle x - \frac{L}{2} \rangle^3 - \frac{1}{8}L^2 x + \frac{5}{48}L^3)$$

When $x = L$, $v(\uparrow) = -\frac{5}{48} \frac{P}{EI}$

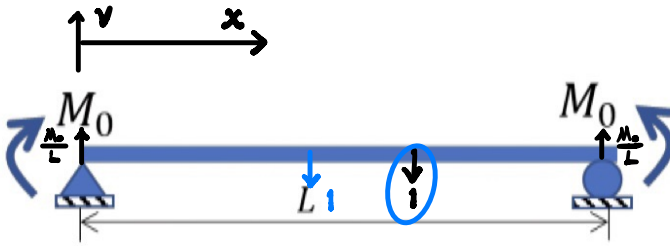
$$\textcircled{2} \quad \delta = \int_0^L \frac{-P \langle x - \frac{L}{2} \rangle}{EI} (-x) dx$$

$$\therefore \delta = \frac{P}{EI} \int_0^{L/2} 0 dx + \frac{P}{EI} \int_{L/2}^L (x^2 - \frac{1}{2}Lx) dx$$

$$= \frac{P}{EI} \int_{L/2}^L (x^2 - \frac{1}{2}Lx) dx = \frac{P}{EI} [\frac{1}{3}x^3 - \frac{1}{4}Lx^2]_{L/2}^L$$

$$= \frac{5}{48} \frac{P}{EI}$$

2. Calculate the maximum deflection using both double integration and unit load approaches. Sketch the deflected shape.



b) $M = \frac{M_0}{L} x$

$$\textcircled{1} \frac{d^2v}{dx^2} = -\frac{1}{R} = -\frac{M}{EI} = -\frac{M_0}{EI} \frac{x}{L}$$

$$\therefore \frac{dv}{dx} = \int -\frac{M_0}{EI} \frac{x}{L} dx = -\frac{M_0}{EI} \left(\frac{x^2}{2L} + C_1 \right)$$

$$\therefore v = -\frac{M_0}{EI} \int \left(\frac{x^2}{2L} + C_1 \right) dx = -\frac{M_0}{EI} \left(\frac{x^3}{6L} + C_1 x + C_2 \right)$$

$$\therefore x=0 \text{ and } L, v=0$$

$$\therefore C_2 = 0$$

$$\therefore C_1 = -\frac{1}{6L} - \frac{1}{2L}$$

$$\therefore v = -\frac{M_0}{EI} \left(\frac{x^3}{6L} - \frac{1}{6}x \right)$$

$$\frac{dv}{dx} = -\frac{M_0}{EI} \left(\frac{x^2}{2L} - \frac{1}{6} \right)$$

When $\frac{dv}{dx} = 0$, deflection goes maximum ($0 < x < L$)

$$\therefore x = \frac{\sqrt{3}}{3}L - \frac{1}{2}L$$

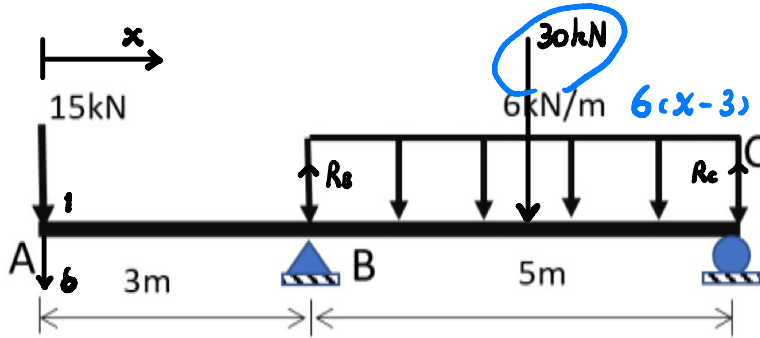
$$\therefore v_{\max} = \frac{\sqrt{3}}{27}L^2 \frac{M_0}{EI} - \frac{1}{8}L^2 \frac{M_0}{EI}$$

$$\textcircled{2} \bar{I} \cdot \delta = \int_0^L \frac{M_0}{EI} \left(\frac{\sqrt{3}}{3}L - \left(x - \frac{1}{3}L \right) \right) dx$$

$$\therefore \delta = \frac{\sqrt{3}}{27}L^2 \frac{M_0}{EI}$$

$$\bar{I} \cdot \delta = \int_0^L \frac{M_0}{EI} \left(\frac{x}{2} - \left(x - \frac{1}{2} \right) \right) dx = -\frac{1}{8}L^2 \frac{M_0}{EI}$$

3. Calculate the at deflection at A using the unit load approaches.



$$R_B = 39 \text{ kN}, R_C = 6 \text{ kN}$$

$$M = -15x + 39\langle x-3 \rangle - 30\langle x-5.5 \rangle$$

When $\bar{1}$ applied to A.

$$R_B = \frac{8}{5}$$

$$\bar{M} = -x + \frac{8}{5}\langle x-3 \rangle$$

$$\therefore \bar{1} \cdot \delta = \int_0^8 \frac{M}{EI} \bar{M} dx$$

$$= \frac{1}{EI} \left(\int_0^3 (-15x - x) dx + \int_3^{5.5} [-15x + 39\langle x-3 \rangle] [-x + \frac{8}{5}\langle x-3 \rangle] dx \right.$$

$$\left. + \int_{5.5}^8 [-15x + 39\langle x-3 \rangle - 30\langle x-5.5 \rangle] [-x + \frac{8}{5}\langle x-3 \rangle] dx \right)$$

$$= \frac{1}{EI} \left(\int_0^3 -16x dx + \int_3^{5.5} \left(\frac{72}{5}x - \frac{927}{5}x + \frac{2808}{5} \right) dx + \int_{5.5}^8 \left(-\frac{18}{5}x + \frac{288}{5}x - \frac{1152}{5} \right) dx \right)$$

$$= \frac{1}{EI} \left(-128 + \frac{525}{4} - \frac{75}{4} \right)$$

$$= -\frac{405}{8EI} - \frac{1065}{4EI}$$

$$\therefore \delta = -\frac{405}{8EI} - \frac{1065}{4EI}$$